# /loop

Goal-driven agent loop: /loop opens setup, /loop starts, /loop help shows usage

## Overview

`/loop` runs a goal-driven agentic loop that drives the selected agent toward an explicit goal until a stop condition is reached. A run stops when the goal is achieved, the verification gate decides the outcome, the step or cost budget is exhausted, or the run is interrupted.

The command has two surfaces: the interactive setup dialog (opened by bare `/loop`) and the one-shot command form that accepts an agent, optional flags, and the goal in a single line. Flags may appear in any position and accept both `--flag value` and `--flag=value` spellings.

## Syntax

```
/loop
/loop help
/loop <agent> <goal>
/loop <agent> [--max-steps N] [--cost_limit N] [--timeout N] <goal>
```

Flag rules:

- Flags may appear anywhere in the line, before or after the goal text.
- `--flag value` and `--flag=value` are both accepted.
- `--cost_limit` and `--cost-limit` are both accepted spellings.
- Values must be non-negative integers.

## Options

| Form | Description |
|---|---|
| `/loop` (bare) | Opens the interactive setup dialog; no run is started. |
| `help` | Prints the usage help. |
| `<agent>` | Agent preset that runs the loop (e.g. `coder`, `general`). |
| `<goal>` | The goal text the loop works toward. Required. |
| `--max-steps N` | Step budget override for this run (non-negative integer). |
| `--cost_limit N` | Cost budget override (alias `--cost-limit`). |
| `--timeout N` | Run timeout in seconds (non-negative integer). |

Defaults when flags are omitted:

| Setting | Default |
|---|---|
| Max steps | `loop.max_steps` from config (512) |
| Cost limit | No cost limit |
| Checkpoints | Enabled |

## Setup dialog

Bare `/loop` opens the setup dialog with these fields:

| Field | Purpose |
|---|---|
| Agent | Agent preset used for the run |
| Goal | The goal text the loop works toward |
| Verify cmd | Optional verification command run by the verification gate |
| Scope | Scope globs restricting where the loop may operate |
| Read-only | Read-only globs the loop may read but not modify |
| Tool set | Restricts the tool set available to the loop |
| Max steps | Step budget override |
| Cost limit | Cost budget override |
| Checkpoints | Toggles pre-loop snapshot capture on or off |

Dialog keys:

| Key | Action |
|---|---|
| Tab / Shift+Tab | Move between fields |
| Up / Down | Move between fields |
| Space | Toggle the Checkpoints field |
| Enter | Confirm and start the loop |

If the goal is missing when a run is requested, the loop reports an error and re-opens the dialog with the agent pre-selected.

## Examples

```
/loop
```

Opens the setup dialog.

```
/loop coder Fix the failing unit tests in crates/ragent-storage
```

```
/loop coder --max-steps 64 Refactor parse_flags into smaller functions
```

```
/loop coder --max-steps=100 --cost_limit=5 --timeout=1800 Make cargo test green
```

```
/loop debug --cost-limit 2 Diagnose the intermittent TUI scroll panic
```

```
/loop help
```

## Output

- On start the loop reports the goal, agent, and any active overrides, then runs step-by-step with the budget enforced.
- Stop conditions: `completed`, `error`, `budget_exhausted`, or `interrupted`.
- With checkpoints enabled, a pre-loop snapshot is captured before the first step; after the loop the rollback flow offers restoring it (Enter restores the pre-loop snapshot, Esc keeps the changes).

- With a verify command configured, the verification gate runs after the loop and its result participates in the stop decision.

## Configuration

| Config key | Default | Purpose |
|---|---|---|
| `loop.max_steps` | 512 | Default step budget when no `--max-steps` flag is given |
| `loop.cost_limit` | none | Default cost budget when no `--cost_limit` flag is given |
| `loop.checkpoint_timeout_secs` | 120 | Checkpoint timeout in seconds |

## Related

- `/autopilot` – autonomous operation with auto-approved permissions
- `/goal` – set a goal condition evaluated after each turn
- `/resume` – continue a halted agent run
- `/undo` – snapshot rollback for file edits
- `ragent.json` `loop` section – loop defaults and checkpoint timeout